TCS NQT · ADVANCED REASONING

Advanced Reasoning Zero to Hero

Every reasoning type from basics to hardest puzzle: method, diagram, formulas, shortcuts, solved examples and graded practice.

13	69	125	42
chapters, zero to hero	solved examples	practice questions	diagrams and charts

ONLINE LEARNING MOCK TEST SERIES · ATC DEVELOPERS

About this guide

This guide covers every reasoning topic that can appear in the TCS NQT Advanced section: seating, puzzles, syllogism, inequalities, data sufficiency, relations, directions, logic, number puzzles, calendar, clock, cubes, series and caselets. Every puzzle in it has a unique answer confirmed by a program that tried all possibilities.

How each chapter is built

- **Idea** in plain words, then a diagram.
- **Formulas and rules** in the yellow box.
- **How the questions appear** in the exam.
- **Solved examples** with full steps.
- **Shortcuts and mistakes** to avoid.
- **Practice set** (Basic, Medium, Advanced) with answers at the end.

How to study

- Study one chapter a day; draw every example yourself before reading the solution.
- Do the Basic questions after each chapter, the Medium the next day.
- Puzzles (chapters 2, 3, 10) are won by practice: repeat them until the method is automatic.
- Finish with the caselet and the 25-minute mini test.

Contents

1. How the Advanced Reasoning Section Works — Foundation — 3
2. Seating Arrangements — Must-do — 5
3. Multi-Attribute Puzzles — Must-do — 8
4. Tournaments, Ranking, Ordering — Important — 11
5. Syllogism (Advanced) — Must-do — 14
6. Inequalities: Direct and Coded — Must-do — 17
7. Data Sufficiency — Important — 19
8. Blood Relations and Directions — Must-do — 22
9. Truth-Tellers and Critical Reasoning — Important — 26
10. Cryptarithm and Number Puzzles — Advanced — 29
11. Calendar, Clock, Cubes, Dice — Important — 33
12. Input-Output, Coding, Series — Must-do — 37
13. Caselets, Mini Test, Final Sheet — Final — 40

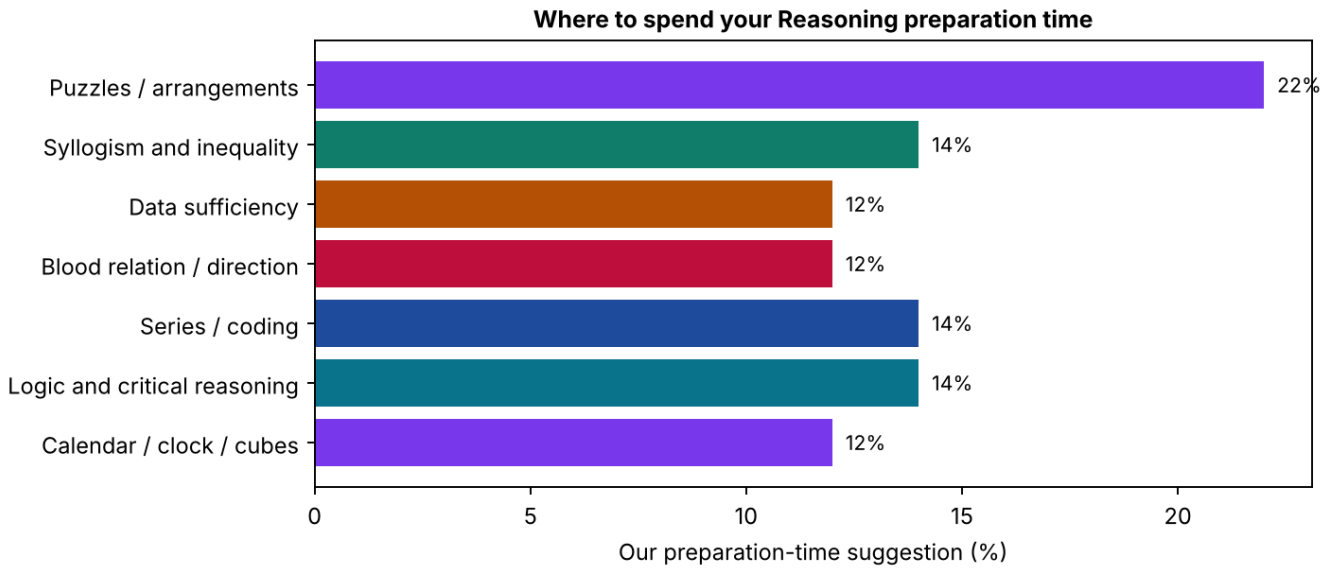
Honest note

- Every number in the solved examples and in the practice answers was calculated by a program while this guide was made, and puzzle answers were found by exhaustive search where possible.
- TCS does not publish its question weightage. Priority labels are our preparation guidance.

ADVANCED REASONING · START HERE

01 How the Advanced Reasoning Section Works

Advanced Reasoning in TCS NQT tests whether you can **hold several conditions in your head at once** and extract one fixed answer. It is not about speed alone: puzzle-style questions reward a method (a grid, a diagram, a table) rather than guesswork. This guide gives you the method for every question type, with a diagram for each.

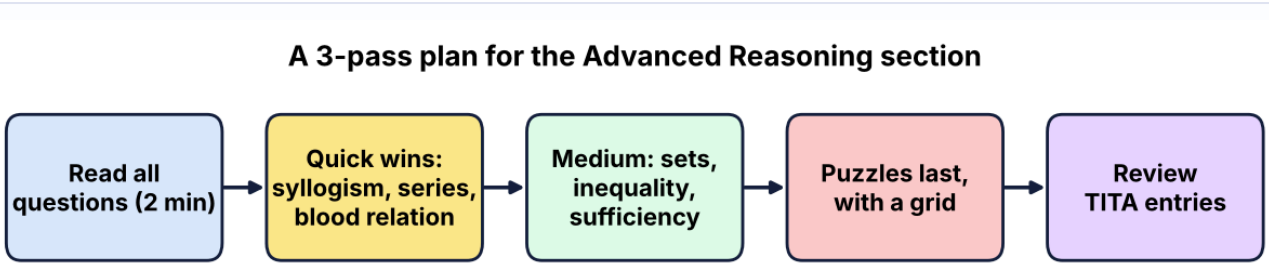


Our guidance only. TCS does not publish a weightage.

The six habits of a high scorer

Habits

- **Draw, never imagine.** Every arrangement question gets a diagram; every “who is related to whom” gets a tree.
- **Start from the most definite statement** (a fixed position, a fixed number, “only one”), not from the first line.
- **Use the negative clues last.** “X is not next to Y” only helps after the positive clues are placed.
- **Make cases when stuck.** Two cases, each short, are faster than one long guess.
- **Always verify the final arrangement against every statement.** This catches the common one-clue misread.
- **TITA means no options:** check the answer against the conditions once more before typing it.



Take the quick questions first; never fight a puzzle in the first minutes.

Question types in this guide

Type	What you do	Chapter
Seating (linear and circular)	Place people with conditions	2
Multi-attribute puzzles	Floors, days, months, professions in a grid	3
Tournament and ranking	Counting matches and positions	4
Syllogism (advanced)	Venn diagrams for all/some/no	5
Inequalities	Coded symbols and chains	6
Data sufficiency	Decide if the data is enough	7
Blood relation and direction	Trees and coordinate paths	8
Truth / liar and critical reasoning	Test the cases; find the flaw	9
Cryptarithm and number puzzles	Digits, columns, grids	10
Calendar, clock, cubes, dice	Formulas and counting	11
Input–output, coding, series	Find the rule	12
Caselets and mini test	Mixed practice	13

What costs marks

- Reading the conditions once and starting to place people at once.
- Skipping the “facing the centre / outside” detail in circular arrangements.
- Over-spending time on one puzzle that is worth a single mark.

Practice set: Warm-up

1. If A > B and B > C, who is the smallest? Basic

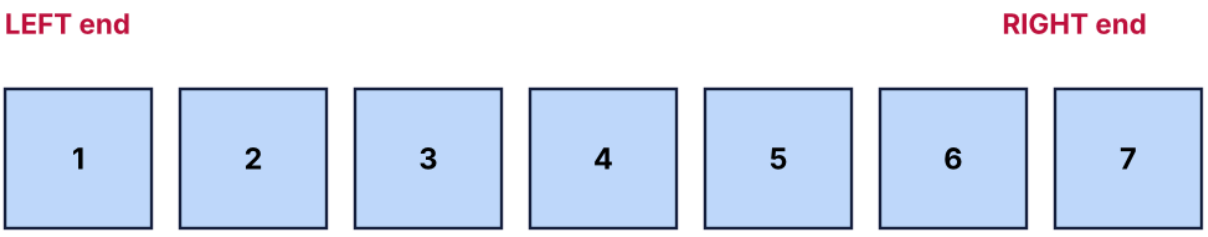
2. Which letter comes next: A, C, E, G, ? Basic

3. Five people sit in a row. How many different orders are possible? Basic

ADVANCED REASONING · SEATING · MUST-DO

02 Seating Arrangements: Linear and Circular

Seating questions are the most common puzzle in TCS NQT Reasoning. They are easy when you follow a fixed method: **(1) draw the empty seats, (2) fix the most definite person, (3) place others from positive clues, (4) use negative clues to eliminate, (5) verify**. The only trap is the direction of left and right.



Facing NORTH: the person's left is the left of the picture. Facing SOUTH it is reversed.

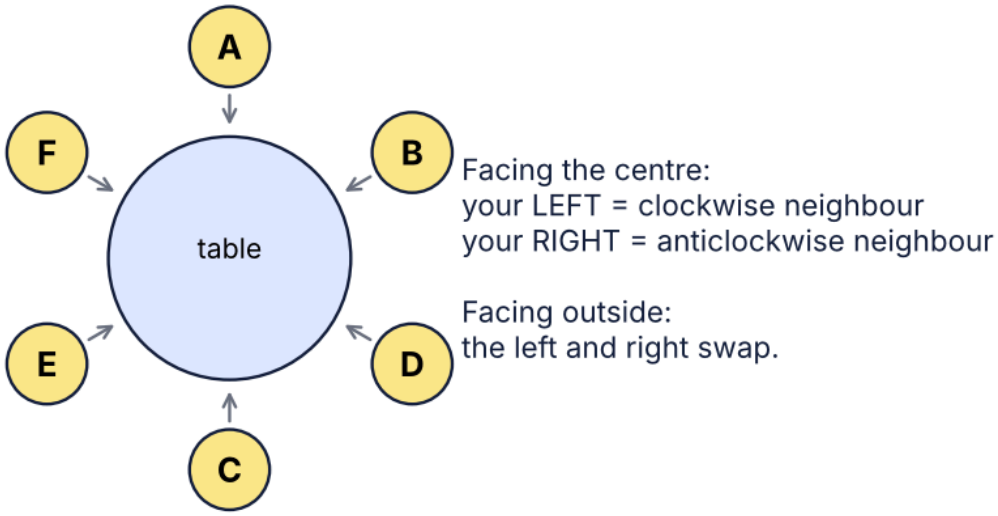
A linear arrangement. Always note which way the people face.

Rules to memorise

- **Facing the centre (circle):** your immediate left is the next person clockwise; your immediate right is anticlockwise.
- **Facing outside:** left and right swap.
- **Opposite in a circle of n people (n even):** position + n/2.
- **Number of arrangements:** n! in a line, (n − 1)! in a circle, 2^(n/2)... only when pairs can swap.
- "Between X and Y" means in the middle; "X is second to the left of Y" means skip one person.
- "Sits next to" = adjacent on either side; "immediate left" = one particular side.
- In a circle with n seats, the distance in either direction is d and n − d.

Worked example: circular table, 6 people

Six people A, B, C, D, E and F sit around a circular table facing the centre. **(1)** B is the immediate left of A. **(2)** E sits opposite B. **(3)** C is not adjacent to A. **(4)** C is the immediate left of D. **(5)** F sits next to E. Find who sits opposite A, and between whom D sits.



The unique arrangement, found by exhaustive search of all 120 circular orders.

Who sits opposite A, and between which two people does D sit?

C; between B and C

Fix A at the top. B (immediate left of A) is the next clockwise seat. E is opposite B (three seats away). C cannot be next to A, and C is the immediate left of D, so D must sit just anticlockwise of C: the only fit is B → D → C. Remaining F next to E closes the circle. Clockwise order: A → B → D → C → E → F. Opposite A is C. D sits between B and C. I checked all five statements on this arrangement, and a program confirmed that no other arrangement works.

Worked example: a row of seven people

Seven people sit in a row facing north. T is at the extreme left. P sits two places to the left of Q. R and S are adjacent. U sits two places to the right of S. Q is to the left of U.

Find the order from the left.

TPVQSRU

T is first. P is two places left of Q, and U is two places right of S with Q left of U. Trying the placements of the adjacent pair R, S, only one keeps every condition. Order from the left: TPVQSRU. A program tried all 5,040 orders and confirmed it is the only one.

Decision flow for any seating question

Step	Do	Why
1	Count seats, draw them	Prevents mixing positions.
2	Note direction (north / centre / outside)	Left and right depend on it.
3	Place fixed positions (extreme, opposite, “third from left”)	They are the anchors.
4	Join people into blocks (“A is left of B”)	Blocks are easier to slide.
5	Try two cases for “either/or” clues	Short cases beat guessing.
6	Verify every clue	Catches misreads.

Shortcuts and tricks

- Make “blocks”: if A is immediate left of B, write AB and treat it as one unit.
- In a circle always fix one person; this removes rotation.
- Opposite clues in a circle pair up people instantly.
- If two answers fit, look for the clue you have not used.

Common mistakes

- Using left/right as seen from the picture instead of the person.
- Forgetting that “between” excludes the end people.
- Placing two people in one seat because a negative clue was applied first.

Practice set: Seating practice

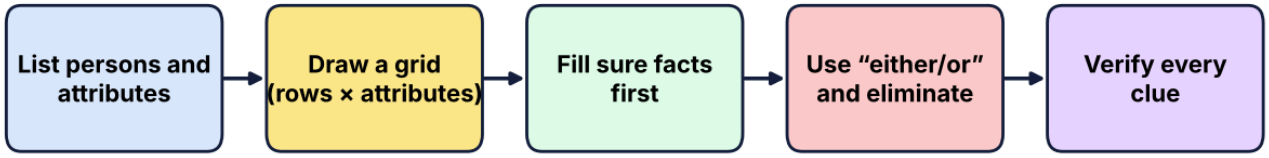
1.	In how many ways can 6 people sit around a circular table?	Basic
2.	In how many ways can 5 people sit in a row?	Basic
3.	In a circle of 8 people, whose position is opposite the person at seat 3?	Basic
4.	In how many ways can 3 men and 3 women sit in a row alternately (men first)?	Medium
5.	In how many ways can 4 men and 4 women sit alternately around a circular table?	Medium
6.	In how many ways can 5 people sit in a row if A and B must be together?	Medium
7.	Seven people sit round a circle. In how many ways can A and B sit together?	Medium
8.	Eight friends sit round a table. In how many arrangements are A and B opposite each other?	Advanced
9.	In the worked circular example, who is the immediate right of A (facing the centre)?	Advanced
10.	Five people sit in a row. A and B must not sit together. How many arrangements?	Advanced

ADVANCED REASONING · PUZZLES · MUST-DO

03 Multi-Attribute Puzzles: Floors, Days, Colours

These puzzles give you 5–8 people and 2–3 attributes (floor, colour, day, profession). The only reliable tool is a **grid**: one row per item, one column per attribute. Fill the sure facts, eliminate with the others, and verify. They look long but each puzzle usually contains **one anchor clue** that cracks the rest.

The grid method for multi-attribute puzzles



The five-step grid method.

Rules

- **Anchor clue:** an exact value ("P lives on floor 3"). Start there.
- **"Immediately above" = +1; "two floors above" = +2; "somewhere above" = any greater floor.**
- **"Between X and Y"** means the middle value, not equal to either.
- **Negative clues** remove options; use them when the grid is almost full.
- **Chain clues:** A > B, B > C ⇒ A > C: write them as a chain first.
- Count: if three people can only take two values, you made a mistake somewhere.

Worked puzzle 1: five floors and five colours

P, Q, R, S and T live on different floors of a five-storey building (floor 1 is the lowest). Each likes a different colour among Red, Blue, Green, Yellow and White. **(1)** P lives on floor 3 and likes Green. **(2)** T lives on floor 1 and likes Red. **(3)** Q lives immediately above R. **(4)** The person who likes Blue lives two floors above S. **(5)** The Yellow lover lives higher than the White lover.

Final grid for the floor puzzle

Floor	Person	Colour
5	Q	Yellow
4	R	Blue
3	P	Green
2	S	White
1	T	Red

The unique solution, found by checking all 14,400 floor and colour combinations.

Who lives on floor 5 and what colour does that person like?

Q; Yellow

P is on 3 and T on 1, so Q, R, S take 2, 4, 5. Q is immediately above R: R = 4, Q = 5, leaving S = 2. Blue is two above S = floor 4, so R likes Blue. Yellow and White remain for floors 2 (S) and 5 (Q); Yellow is higher, so Q likes Yellow and S likes White.

Which person lives immediately below P, and which colour does that person like?

S; White

Floor 2 is S, who likes White.

Worked puzzle 2: five days

Five friends A, B, C, D and E each visit a museum on a different day from Monday to Friday. **(1)** C visits on Wednesday. **(2)** B visits two days after A. **(3)** E visits before D. **(4)** A and D do not visit on Monday.

On which day does each friend visit?

E: Monday, A: Tuesday, C: Wednesday, B: Thursday, D: Friday

C = Wednesday. B is two days after A with A not on Monday: A = Tuesday, B = Thursday (A = Monday is forbidden; A = Wednesday clashes with C). The remaining days are Monday and Friday for D and E. D cannot be Monday and E is before D, so E = Monday, D = Friday. A program tried all 120 orders; this is the only one.

Type	Example	Key idea
Floors / positions	“Q lives immediately above R”	Differences of +1, +2.
Days / months	“B visits two days after A”	Position arithmetic; wrap around week.
Professions	“The doctor does not like red”	Grid of ✔ and ✘.
Counting	“How many people live above S?”	Final grid, then count.
Conditional	“If P is on 4 then …”	Branch cases.

Shortcuts and tricks

- Mark every cell ✔ or ✘ rather than writing nothing: you see contradictions early.
- Positive clues with exact values come first; “not” clues come last.
- When two persons can swap, a later clue is always waiting to break the tie.
- Redraw the grid after each major deduction to keep it readable.

Common mistakes

- Using “above” as “immediately above”.
- Forgetting that “between” is strict.
- Not checking the very first clue again at the end.

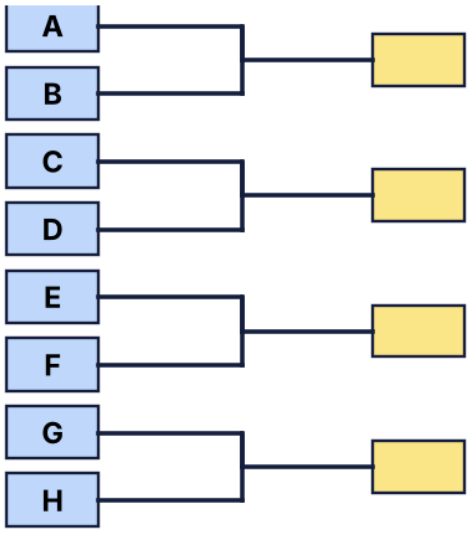
Practice set: Puzzle practice

1. Floors 1 to 5: the sum of floor numbers of all five persons? **Basic**
2. In puzzle 1, on which floor does the Blue lover live? **Basic**
3. In puzzle 1, who lives on floor 2? **Basic**

4.	In puzzle 2, which friend visits on Tuesday?	Medium
5.	In puzzle 2, who visits the day after C?	Medium
6.	In puzzle 1, how many people live above P?	Medium
7.	In puzzle 2, how many friends visit before B?	Medium
8.	In puzzle 1, who likes White?	Advanced
9.	How many different arrangements of 5 people into 5 floors exist?	Advanced
10.	Puzzle 1 has how many solutions in total (found by program)?	Advanced

04 ADVANCED REASONING · RANKING · IMPORTANT
Tournaments, Ranking and Ordering

Ranking questions use positions in a line (rank from top and rank from bottom). Tournament questions use counting formulas. Both are short once you know the three identities below.



Knockout: 8 teams → 7 matches (n − 1)

Rounds = $\log_2 8 = 3$
League (round robin): n(n−1)/2 = 28
Double league: n(n−1) = 56
Teams in round k from the end: 2^k

Knockout and league counts.

Ranking identities

- **Total = rank from left + rank from right − 1.**
- **People between X and Y** (X left of Y) = Y's position − X's position − 1.
- **Swapping positions:** if X and Y exchange places, the new rank of X from left = old rank of Y from left.
- **Position from the other end** = Total + 1 − position.
- **Counting people who are above X** = rank of X − 1.

Tournament formulas

- **Knockout:** n teams → n − 1 matches; rounds = $\lceil \log_2 n \rceil$; byes if n is not a power of 2.
- **League (single round robin):** n(n−1)/2 matches; each team plays n − 1.
- **Double round robin:** n(n−1).
- **Groups:** g groups of k teams → g·k(k−1)/2 group matches.
- **Points tables:** total points = matches × points per match.

In a row of 40 students, A is 12th from the left and B is 18th from the right. How many students are between them?

10

B's position from the left = 40 + 1 − 18 = 23. People between = 23 − 12 − 1 = 10.

In a class, a student is 12th from the top and 25th from the bottom. How many students are in the class?

36

Total = 12 + 25 − 1 = 36.

Two students swap places: A moves from 9th to 14th from the left and B moves from 14th to 9th. If the class has 30 students, what is the position of A from the right?

17

A's new rank from the left is 14. From the right = 30 + 1 − 14 = 17.

A knockout tournament has 37 teams. How many matches are played?

36

Each match removes exactly one team, and 36 teams must be removed to leave one champion: 36 matches. (Byes do not count as matches.)

In a league with 12 teams, each team plays every other team twice. How many matches?

132

Double round robin: n(n−1) = 12 × 11 = 132.

In a cricket league each of 9 teams plays 8 matches; a win is 2 points, a tie 1, a loss 0. How many total points in the table?

72

Matches = 9×8/2 = 36; each gives 2 points in total. Total = 72.

Among 5 people A–E: A is taller than B, B is taller than C, D is taller than A but shorter than E. Who is the shortest and who is the second tallest?

C; D

Chain: E > D > A > B > C. Shortest is C, second tallest D.

Six people have different heights. How many pair-wise comparisons are needed in the worst case to find the tallest?

5

Each comparison eliminates one person from the tallest contest: n − 1 = 5.

Pattern	Wording	Formula
Ranks both ends	"12th from top, 25th from bottom"	12 + 25 − 1
People between	"how many are between A and B"	$\lvert pos_1 - pos_2 \rvert - 1$
Swaps	"if they swap ..."	New rank = other's old rank
Knockout	"37 teams, how many matches"	n − 1
League	"every team plays every other"	n(n−1)/2
Chain order	"A > B, B > C, D < A"	Draw a chain

Shortcuts and tricks

- Draw a line and mark known positions; use the numbers, not the words.
- If “top 5” and “bottom 6” overlap, total is at most 10.
- When a rank is counted from both ends, the formula never changes.

Common mistakes

- Forgetting the −1 in the total (the person is counted twice).
- Counting a bye as a match.
- Mixing left-to-right with right-to-left ranks.

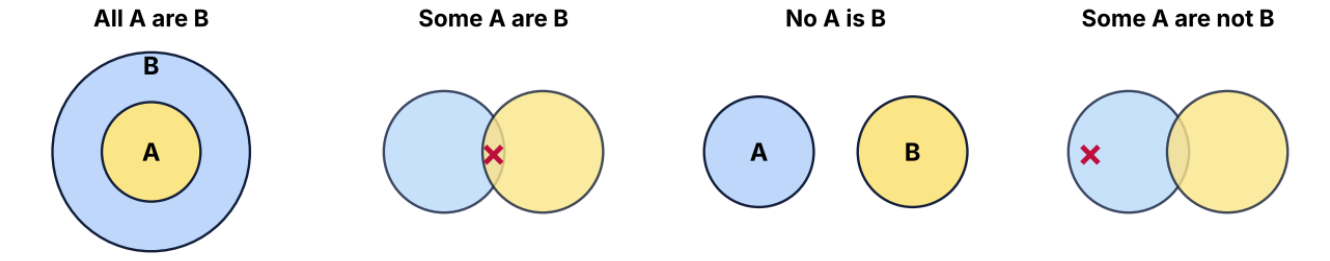
Practice set: Ranking practice

1.	A is 5th from the left and 9th from the right in a row. How many are in the row?	Basic
2.	A knockout tournament has 64 teams. How many matches?	Basic
3.	How many matches in a single league of 10 teams?	Basic
4.	In a row of 35, M is 9th from the left. What is the rank from the right?	Medium
5.	In a row of 45 students, X is 15th from the left and Y is 20th from the right. How many are between them?	Medium
6.	A tournament has 100 teams (knockout). How many matches?	Medium
7.	Four groups of 5 teams each play a league within the group. How many matches?	Medium
8.	A class has 50 students. Rahul is 15th from the top and Sumit is 18th from the bottom. How many students are between them?	Advanced
9.	A and B swap places. A was 7th from the left, B was 15th from the left; the row has 30. A’s new rank from the right?	Advanced
10.	A league of n teams has 45 matches. Find n.	Advanced

ADVANCED REASONING · SYLLOGISM · MUST-DO

05 Syllogism (Advanced)

Syllogism asks: “given these statements, which conclusions **definitely** follow?” The only safe method is the Venn diagram: draw every possible diagram that fits the statements, and accept a conclusion only if it is true in **all** of them. Every answer in this chapter was checked by a program that tests all 255 possible Venn-region models.



The four statement types as diagrams.

The four statement types

- **All A are B** → A is fully inside B. Reverse is “Some B are A” (when A exists).
- **Some A are B** → an overlap exists. Reverse: “Some B are A”.
- **No A is B** → separate circles. Reverse: “No B is A”.
- **Some A are not B** → part of A lies outside B. No safe reverse.
- Valid inferences: All + All = All; All + Some = Some; Some + All = Some; All + No = No (A in B, no B is C ⇒ no A is C).
- Two “Some” statements (“Some A are B, Some B are C”) give **no** conclusion.
- Two “No” statements give no conclusion.
- **Either–or:** if two conclusions form a complementary pair (“Some A are B” / “No A is B”) and neither follows alone, then “either … or” follows.

Statements: All X are Y. All Y are Z. Conclusions: (1) All X are Z (2) Some Z are X (3) Some Z are not X (4) No Z is X. Which follow?

(1) follows, (2) follows, (3) does not follow, (4) does not follow

X ⊂ Y ⊂ Z, so (1) is true. If all X are Z, then some Z are X (reverse). “Some Z are not X” is possible but not certain (X and Z could be equal), and “No Z is X” is false.

Statements: Some X are Y. All Y are Z. Conclusions: (1) Some X are Z (2) Some Z are X (3) All X are Z (4) No X is Z.

(1) follows, (2) follows, (3) does not follow, (4) does not follow

The X that are Y are inside Z, so some X are Z and the reverse holds. Not all X need be Z; “No X is Z” is false.

Statements: Some X are Y. Some Y are Z. Conclusions: (1) Some X are Z (2) No X is Z (3) Either (1) or (2) follows.

(1) does not follow, (2) does not follow; (3) follows

Two "Some" statements give no certain conclusion. But exactly one of "Some X are Z" and "No X is Z" must be true in any diagram, so the either–or conclusion (3) follows.

Statements: All X are Y. No Y is Z. Conclusions: (1) No X is Z (2) Some X are not Z (3) No Z is X (4) Some Y are not Z.

(1) follows, (2) follows, (3) follows, (4) follows

X is inside Y and Y is separate from Z, so X and Z do not overlap: (1) and (3) follow. (2) "Some X are not Z" also follows since X is non-empty and all of it lies outside Z. (4) also follows.

Statements: Some X are Y. No Y is Z. Conclusions: (1) Some X are not Z (2) No X is Z (3) Some Z are not X.

(1) follows, (2) does not follow, (3) does not follow

The X that are Y are outside Z, so some X are not Z. "No X is Z" is not certain because other X could be Z. "Some Z are not X" is not certain: all of Z could sit inside X, in the part of X that is outside Y.

Pair	Conclusion that follows	Typical trap
All–All	All (and Some, reverse)	Reverse "All" is not valid.
All–Some	Some	Not "All".
Some–All	Some	Reverse of Some is fine.
Some–Some	Nothing (maybe either–or)	Do not join two Somes.
All–No	No	Link only through the middle term.
Some–No	Some ... not	"Some A are not C".
No–All	Some ... not	"Some C are not A".
No–No	Nothing	Possible overlap.

Shortcuts and tricks

- Make the "All" diagram first: it fixes the circles; then add "Some" as ×.
- Test each conclusion in its worst diagram: if one diagram breaks it, it does not follow.
- Possibility questions ("can be true") need only one diagram.
- A conclusion with "Some ... not" follows only when part of the set is surely outside.

Common mistakes

- Treating "Some A are B" as "All A are B".
- Joining two particular statements.
- Accepting a conclusion that is merely possible.

Practice set: Syllogism practice (answer: follows / does not follow)

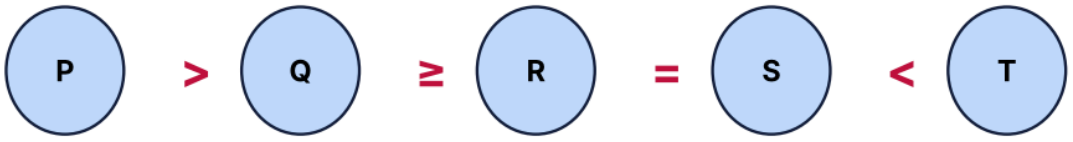
1. Statements: All X are Y. Some Y are Z. Does "Some X are Z" follow? **Basic**
2. Statements: All X are Y. All Y are Z. Does "Some Z are X" follow? **Basic**
3. Statements: No X is Y. All Z are X. Does "No Z is Y" follow? **Medium**

4.	Statements: Some X are Y. All Y are Z. Does “All X are Z” follow?	Medium
5.	Statements: All X are Y. No Z is Y. Does “No X is Z” follow?	Medium
6.	Statements: Some X are Y. Some Y are Z. Does “Some X are Z” follow?	Medium
7.	Statements: All X are Y. Some Z are not Y. Does “Some Z are not X” follow?	Advanced
8.	Statements: No X is Y. Some Z are Y. Does “Some Z are not X” follow?	Advanced
9.	Statements: All X are Y. All Z are Y. Does “Some X are Z” follow?	Advanced
10.	Statements: Some X are not Y. All Y are Z. Does “Some X are not Z” follow?	Advanced

06 Inequalities: Direct and Coded

ADVANCED REASONING · INEQUALITY · MUST-DO

An inequality question gives a chain (or several chains) and asks which conclusion is **definitely true**. The tools: (1) a chain only connects ends when every link points the same way, (2) "=" can be dropped into either direction, (3) a conclusion with "≥ or ≤" is true if the chain gives a strict or equal link.



P > Q ≥ R = S < T : P > R and P > S are definite; P vs T is uncertain.

Strict links > and < , loose links ≥ and ≤ , equal =.

Rules

- **Same direction chain:** A > B ≥ C ⇒ A > C (strict + loose = strict). A ≥ B ≥ C ⇒ A ≥ C.
- **Mixed directions give no relation:** A > B < C says nothing about A and C.
- **Equal sign:** A = B ≥ C ⇒ A ≥ C.
- **Either–or conclusions:** if "A > C" and "A = C" both fail separately but together cover cases, the answer is "either ... or". For ≥ the complete pair is "A > C" or "A = C".
- **Coded symbols:** first decode (e.g. @ = >, # = <), then solve as a normal chain.

P > Q ≥ R = S < T. Which are definitely true? (1) P > R (2) P > S (3) P > T (4) R < T (5) Q ≥ S

(1) true, (2) true, (3) uncertain, (4) true, (5) true

P > Q ≥ R makes P > R. R = S gives P > S. R = S < T gives R < T. P and T are connected only by opposite directions (P > ... < T), so no relation. Q ≥ R = S gives Q ≥ S. I verified each by trying every ordering of the five letters.

A ≥ B > C and D = B. Which hold? (1) A > C (2) D > C (3) A ≥ D (4) D ≥ A

(1) true, (2) true, (3) true, (4) uncertain

A ≥ B > C gives A > C. D = B > C gives D > C. A ≥ B = D gives A ≥ D. D ≥ A would need A = B, which is not certain.

Coded inequality

Symbols: **@** means "greater than", **#** means "less than", **$** means "greater than or equal to", **%** means "less than or equal to", ***** means "equal to".

Statement: M @ N $ O # P. Conclusions: (1) M > O (2) N > P (3) M > P (4) P > O

(1) true, (2) uncertain, (3) uncertain, (4) true

Decode: M > N ≥ O < P. (1) M > N ≥ O gives M > O. (2) N and P are linked through the opposite directions N ≥ O < P, so unknown. (3) M and P are connected via a valley, unknown. (4) O < P means P > O.

Either–or and “at least one”

X ≥ Y > Z. Conclusions: (1) X > Z (2) X = Z (3) Z > X (4) Z ≥ X

(1) true, (2) false, (3) false, (4) false

X ≥ Y > Z means X > Z in every case, so only (1) is true.

Chain	Gives	Reason
A > B > C	A > C	Same direction
A ≥ B > C	A > C	One strict link
A ≥ B ≥ C	A ≥ C	Loose
A > B < C	—	Opposite
A = B > C	A > C	Equal passes through
A ≥ B, B ≥ A	A = B	Both sides

Shortcuts and tricks

- Rewrite the whole statement as one chain before reading any conclusion.
- If two letters are linked only by a “valley” (>...<) or a “hill” (<...>), no conclusion.
- For “either ... or” the two conclusions must be complementary (> and =, or < and =).

Common mistakes

- Combining A ≥ B ≤ C.
- Dropping the “or equal” and calling a conclusion false.
- Misreading the coded symbol.

Practice set: Inequality practice (true / false / uncertain)

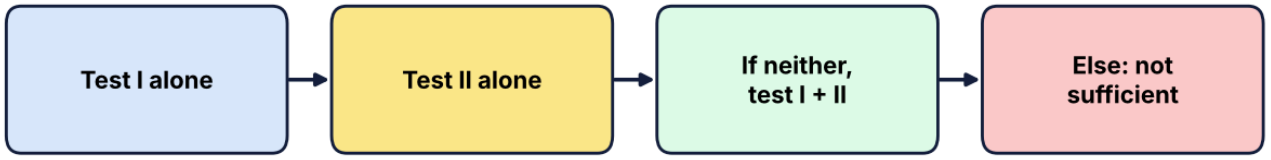
1. P > Q > R. Is P > R true? — Basic
2. A ≥ B ≥ C. Is A > C definite? — Basic
3. A > B < C. Is A > C definite? — Basic
4. K ≥ L = M > N. Is K > N definite? — Medium
5. D < E ≤ F = G. Is D < G definite? — Medium
6. H > I ≥ J, J = K. Is H > K definite? — Medium
7. Decode and test: A % B * C @ D (@ is >, % is ≤, * is =). Is A < D definite? — Advanced
8. S ≥ T > U, V ≤ T. Is S > V definite? — Advanced
9. W = X, X > Y, Z ≥ Y. Is W > Z definite? — Advanced
10. B ≥ C ≥ D, D ≥ B. Is C = D definite? — Advanced

07 Data Sufficiency

ADVANCED REASONING · DATA SUFFICIENCY · IMPORTANT

Data sufficiency does not ask you for the answer. It asks whether the **given statements are enough to find one single answer**. The classic five options are: (A) Statement I alone, (B) Statement II alone, (C) either alone, (D) both together are needed, (E) not even together. Always test I and II separately before combining.

Data sufficiency: always follow this order



Never combine before testing each statement alone.

Rules

- **A statement is sufficient if it gives exactly one value** (or a definite "yes", or a definite "no" for yes/no questions).
- **For "What is x?"** sufficiency means a single value, not a range.
- **For "Is x > 5?"** a definite "no" is also sufficient. "Sometimes yes, sometimes no" is not.
- **Do not use outside knowledge** unless the question says it is common knowledge.
- **Combine only when needed.** Both together may give an answer even if neither does alone.
- **Linear equations:** two unknowns need two independent equations. Equations that are multiples of each other count as one.

What is x? I: x + y = 10. II: x − y = 2.

Both together are needed

Each statement alone leaves many (x, y) pairs. Together: x = 6, y = 4. I tested all integer pairs from −30 to 30.

What is x? I: 2x + 3y = 12. II: 4x + 6y = 24.

Even both together are not sufficient

II is exactly twice I, so both are the same equation. Even together they leave many solutions.

Is x > y? I: x = 5. II: y = 3.

Both together are needed

Alone, each leaves the other variable free. Together x = 5, y = 3 and the answer is "yes".

Is x even? (x is a positive integer ≤ 30.) I: x is a multiple of 4. II: x is a multiple of 3.

Statement I alone is sufficient

Multiples of 4 are always even, so I alone settles it ("yes"). Multiples of 3 can be odd or even, so II alone does not.

How old is A? (age in whole years 1–100) I: A is 4 years older than B. II: The sum of the ages of A and B is 30.

Both together are needed

Neither alone fixes A. Together A = 17, B = 13.

Is x > 0? I: $x^2 > 4$. II: $x^3 > 0$.

Statement II alone is sufficient

$x^2 > 4$ gives $x > 2$ or $x < -2$, so x can be negative: not sufficient. $x^3 > 0$ forces $x > 0$: sufficient.

Situation	Verdict
One equation, two unknowns	Not sufficient alone
Two unknowns, two independent equations	Sufficient together
Statement is a multiple of the other	Counts as one
Yes/no question, statement always "no"	Sufficient
Statement gives a range	Not sufficient (for "find x")
$x^2 = 9$	$x = \pm 3$, not unique unless $x > 0$ is known

Shortcuts and tricks

- Look for hidden constraints: "positive integer", "prime", "less than 20".
- For yes/no questions, aim to find one example for yes and one for no; if both exist, not sufficient.
- Squares and absolute values hide two answers.

Common mistakes

- Combining before testing each statement.
- Treating "probably yes" as sufficient.
- Missing a negative root.

Practice set: Data sufficiency practice

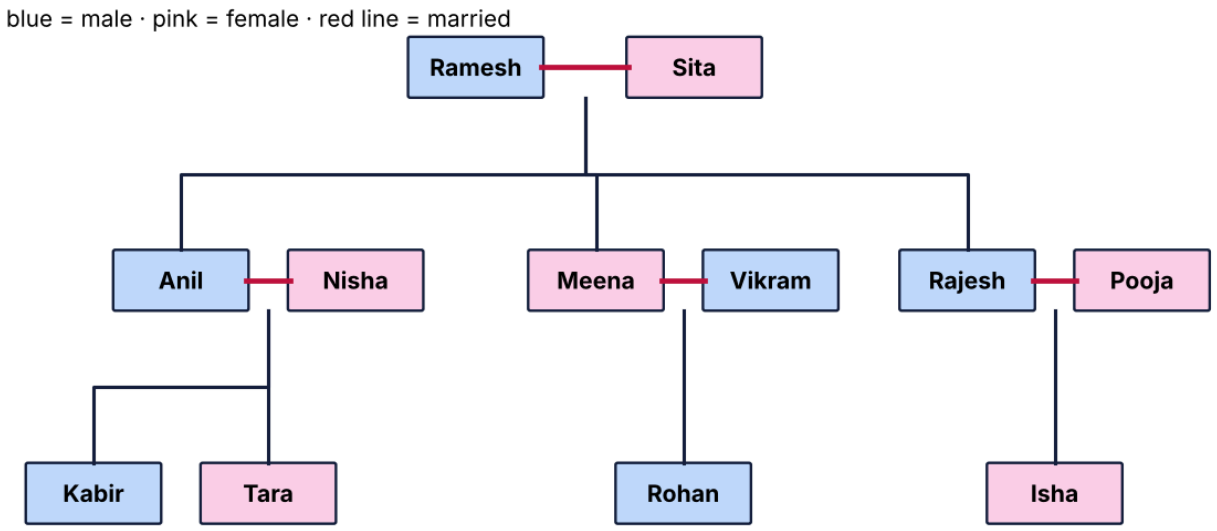
1. What is x? I: $x = 7$. II: $y = 3$. — Basic
2. What is x? I: $x + y = 5$. II: $x - y = 1$. — Basic
3. What is x? I: $x^2 = 25$. II: $x > 0$. — Medium
4. What is x? I: $x + y = 8$. II: $3x + 3y = 24$. — Medium
5. Is $x > y$? I: $x = 2y$. II: $y > 0$. — Medium
6. Is $x + y$ even? I: x is even. II: y is odd. — Medium
7. Is x positive? I: $x^2 > 0$. II: $x^3 > 0$. — Advanced
8. What is $x \cdot y$? (positive integers) I: $x + y = 5$, $x > y$. II: $x = 4$. — Advanced

9.	What is x? (x is a positive integer ≤ 30) I: x is a prime less than 5. II: x is a multiple of 3.	Advanced
10.	Is x an integer multiple of 6? (x positive integer ≤ 30) I: x is a multiple of 2. II: x is a multiple of 3.	Advanced

ADVANCED REASONING · RELATIONS AND DIRECTION · MUST-DO

08 Blood Relations and Directions

Blood-relation questions are decoded with a **family tree**: one symbol per person, a line for marriage, a line down for children. Direction questions are decoded with a **coordinate grid**: north is up, east is right, and right/left turns rotate the facing direction.



One family tree used for every example in this section.

Relation vocabulary

- Father's / mother's brother = **uncle**; father's / mother's sister = **aunt**.
- Uncle's or aunt's child = **cousin**.
- Sister's / brother's son = **nephew**, daughter = **niece**.
- Spouse's brother or sister, or sibling's spouse = **brother/sister-in-law**.
- Son's wife = daughter-in-law; daughter's husband = son-in-law.
- Mother's mother or father's mother = **grandmother**.
- "Only son" and "only daughter" fix the number of siblings.
- **Gender unknown?** Use words like "child" or "sibling" until a clue decides.

Using the tree: what is Rohan to Anil?

nephew

Trace the path from Anil to Rohan on the tree and name the relation. The program applied the same rules to the tree and returned "nephew".

Using the tree: what is Rajesh to Tara?

uncle

Trace the path from Tara to Rajesh on the tree and name the relation. The program applied the same rules to the tree and returned “uncle”.

Using the tree: what is Meena to Kabir?

aunt

Trace the path from Kabir to Meena on the tree and name the relation. The program applied the same rules to the tree and returned “aunt”.

Using the tree: what is Anil to Rohan?

uncle

Trace the path from Rohan to Anil on the tree and name the relation. The program applied the same rules to the tree and returned “uncle”.

Using the tree: what is Sita to Isha?

grandmother

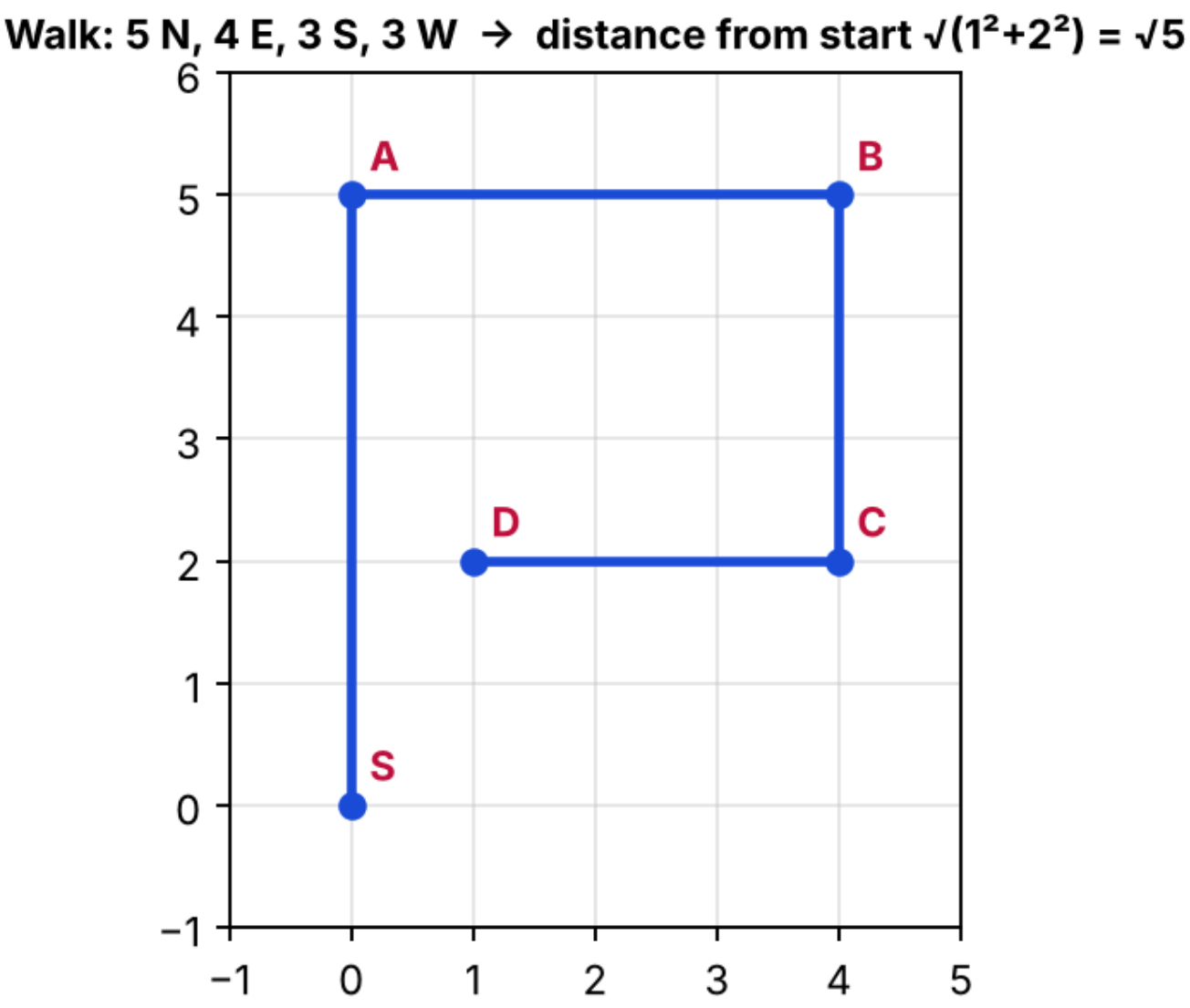
Trace the path from Isha to Sita on the tree and name the relation. The program applied the same rules to the tree and returned “grandmother”.

Pointing to a photo, a man says, “His mother is the only daughter of my mother.” How is the man related to the person in the photo?

Maternal uncle

The man’s mother’s only daughter is his sister. The photo’s person’s mother is the man’s sister, so the man is that person’s maternal uncle.

Directions



Plot the path; the straight-line distance is the diagonal.

A man walks 5 m north, turns right and walks 4 m, turns right and walks 3 m, turns right and walks 3 m. How far is he from the start, and in which direction does he now face?

2.236 m; facing W; position (1,2)

Start at (0,0). North 5 → (0,5); right turn → east 4 → (4,5); right → south 3 → (4,2); right → west 3 → (1,2). Distance = √(1² + 2²) = √5 ≈ 2.236 m. The last move was west, so he faces W.

A person facing east turns 90° clockwise, then 180°, then 90° anticlockwise. Which way is he facing now?

W (facing W)

Clockwise 90° from east = south; 180° more = north; 90° anticlockwise from north = west.

Shadow rule: At sunrise, a man's shadow falls to his right. Which direction is he facing?

South

At sunrise the sun is in the east, so shadows fall west. A shadow on his right means his right is west, so he faces south.

Pattern	Method	Trap
Direct relation	Draw a tree	Gender not given
Coded relation (A + B means ...)	Decode each symbol	Order of symbols
Photograph	Break "his/her" in two steps	Reverse the pointing
Distance after turns	Plot x,y	Using the path length
Shadow	Sun in the east at sunrise	Left/right swap
Compass turns	Right = +90°	Mixing clockwise

Shortcuts and tricks

- Mark each person with M/F (or ? if unknown) as soon as the sentence allows.
- Draw a vertical line for parent→child and a horizontal one for spouse.
- For distance use Pythagoras only on the final displacement, not the path.

Common mistakes

- Assuming gender from a name.
- Counting the walk length instead of the displacement.
- Mixing "my father's sister" (aunt) with "my mother's sister" (aunt also) but forgetting "maternal".

Practice set: Relations and directions practice

1. Using the tree: what is Isha to Rajesh? — Basic
2. What is Meena to Kabir? — Basic
3. A man faces north, turns right, then right again. He now faces? — Basic
4. What is Tara to Rohan? — Medium
5. What is Anil to Isha? — Medium
6. What is Nisha to Rohan? — Medium
7. A walks 6 m east, then 8 m north. Distance from start? — Medium
8. A person walks 10 m west, 5 m south, 6 m east, 5 m north. Where is he relative to the start (west is negative)? — Advanced
9. What is Vikram to Kabir? — Advanced
10. What is Sita to Isha? — Advanced

09 Truth-Tellers, Liars and Critical Reasoning

ADVANCED REASONING · LOGIC · IMPORTANT

Two families of logic questions: **truth/liar puzzles** (try each person as the truth-teller) and **critical reasoning** (strengthen, weaken, find the assumption, find the flaw). The first is solved by a short table of cases; the second by understanding what a conclusion needs.

Only the contrapositive is always valid

If P then Q	Statement	Valid?
Original	If it rains, the road is wet	—
Converse	If the road is wet, it rained	No
Inverse	If it does not rain, the road is not wet	No
Contrapositive	If the road is not wet, it did not rain	Yes

The contrapositive is the only equivalent form of an if-then statement.

Logic rules

- **If P then Q** equals **if not Q then not P** (contrapositive). It does **not** equal "if Q then P".
- **"Only if":** "P only if Q" means if P then Q.
- **"Unless":** "A unless B" means if not B then A.
- **"Either ... or"** in puzzles usually means at least one; "exactly one" means not both.
- **Truth-teller always speaks truth, liar always lies:** if X says "I am a liar" the statement is a paradox, so such a puzzle does not exist.
- **Method:** assume person 1 is truthful, follow the consequences, check for contradiction; try the other case.

A says "B is lying." B says "C is lying." C says "A and B are both lying." Who tells the truth?

B tells the truth

A program tried all 8 truth patterns. Only one is consistent: A=lie, B=truth, C=lie.

Four people say: A "exactly one of us is lying", B "exactly two of us are lying", C "exactly three of us are lying", D "all four of us are lying". Who tells the truth?

C

At most one of these mutually exclusive claims can be true. If exactly three are lying then C is the only truth-teller: C's own claim "exactly three lie" is then true. A program checked all 16 patterns and confirmed this is the only consistent one.

Statement: "If a candidate clears the aptitude round, he is called for the interview." Aman was not called for the interview. What follows?

He did not clear the aptitude round

Contrapositive: not called ⇒ did not clear the aptitude round. (Truth-table check: contrapositive equivalent = True; converse equivalent = False.)

Critical reasoning: the four question types

Type	Question	What to do
Assumption	"The argument assumes ..."	Find the unstated link between premise and conclusion.
Strengthen	"Which most strengthens ..."	Add evidence that makes the conclusion more likely.
Weaken	"Which most weakens ..."	Show an alternative cause or a counter-example.
Flaw	"The reasoning is flawed because ..."	Look for correlation vs causation, small sample, false either-or.
Inference	"Which must be true ..."	Only what follows directly; nothing extra.
Paradox	"Resolve the apparent contradiction"	The option that makes both facts true.

Argument: "After the city opened a new metro line, car sales in the city fell. Therefore the metro caused the fall." Which of the following most weakens the argument? (a) Metro fares are low. (b) Car sales fell in neighbouring cities with no metro. (c) The metro serves 5 lakh people a day. (d) Some people prefer cars.

(b)

If sales fell everywhere, including cities without a metro, the metro is not the cause. The other options do not challenge the causal claim.

Argument: "Company X's ads increased in 2023 and sales increased. So ads increase sales." What is the assumption?

No other factor caused the sales increase

The argument assumes the sales rose because of the ads, not because of price cuts, festivals or other causes.

Shortcuts and tricks

- Weaken = break the link; strengthen = repair the link.
- Be wary of words like "all", "always", "only" in options: they are usually too strong.
- For "must be true" questions, pick the option that is hardest to deny.

Common mistakes

- Using outside knowledge instead of the passage.
- Accepting a converse as valid.
- Treating correlation as proof of cause.

Practice set: Logic practice

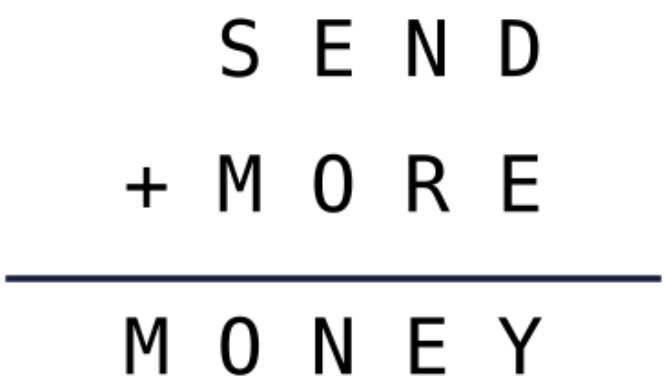
1. "If it rains, the match is cancelled." The match was not cancelled. What follows? **Basic**
2. Is "If the road is wet, it rained" equivalent to the original "If it rains, the road is wet"? **Basic**

3.	Is the contrapositive equivalent to the original?	Basic
4.	A says “B lies.” B says “A lies.” How many people can tell the truth?	Medium
5.	A: “C is a liar.” B: “A is truthful.” C: “B is a liar.” Who is truthful? (try all 8)	Medium
6.	“No manager is a fresher; some fresher are tall.” Can we conclude “some tall persons are not managers”?	Medium
7.	How many truth patterns exist for 5 people?	Advanced
8.	In the 4-person puzzle above, how many are lying?	Advanced
9.	A “only if” B means?	Advanced
10.	“A unless B” is equivalent to?	Advanced

10 Cryptarithm, Number Puzzles and Grids

ADVANCED REASONING · NUMBER PUZZLES · IMPORTANT

Number puzzles replace digits by letters or hide a pattern in a grid. They are won by **carries, parity and divisibility**, not by guessing. Always begin with the leftmost column: a carry from addition is at most 1.



M = 1 (carry) S = 9 O = 0
E = 5 N = 6 D = 7 R = 8 Y = 2

The classic cryptarithm. M must be 1 (carry).

Cryptarithm toolkit

- Adding two 4-digit numbers gives at most 19998, so the carry into the 5th digit is 1.
- Leading digits cannot be 0.
- Different letters represent different digits (unless said otherwise).
- Units column: use the last digit; carry: use ≤ 1 (two numbers) or ≤ 2 (three numbers).
- If a letter appears in the same column in both addends: the sum is even.
- Digit sum of 3-digit and 4-digit numbers is checked by mod 9.

Solve SEND + MORE = MONEY (distinct digits, no leading zero).

1 solution: O=0, M=1, Y=2, E=5, N=6, D=7, R=8, S=9; SEND=9567

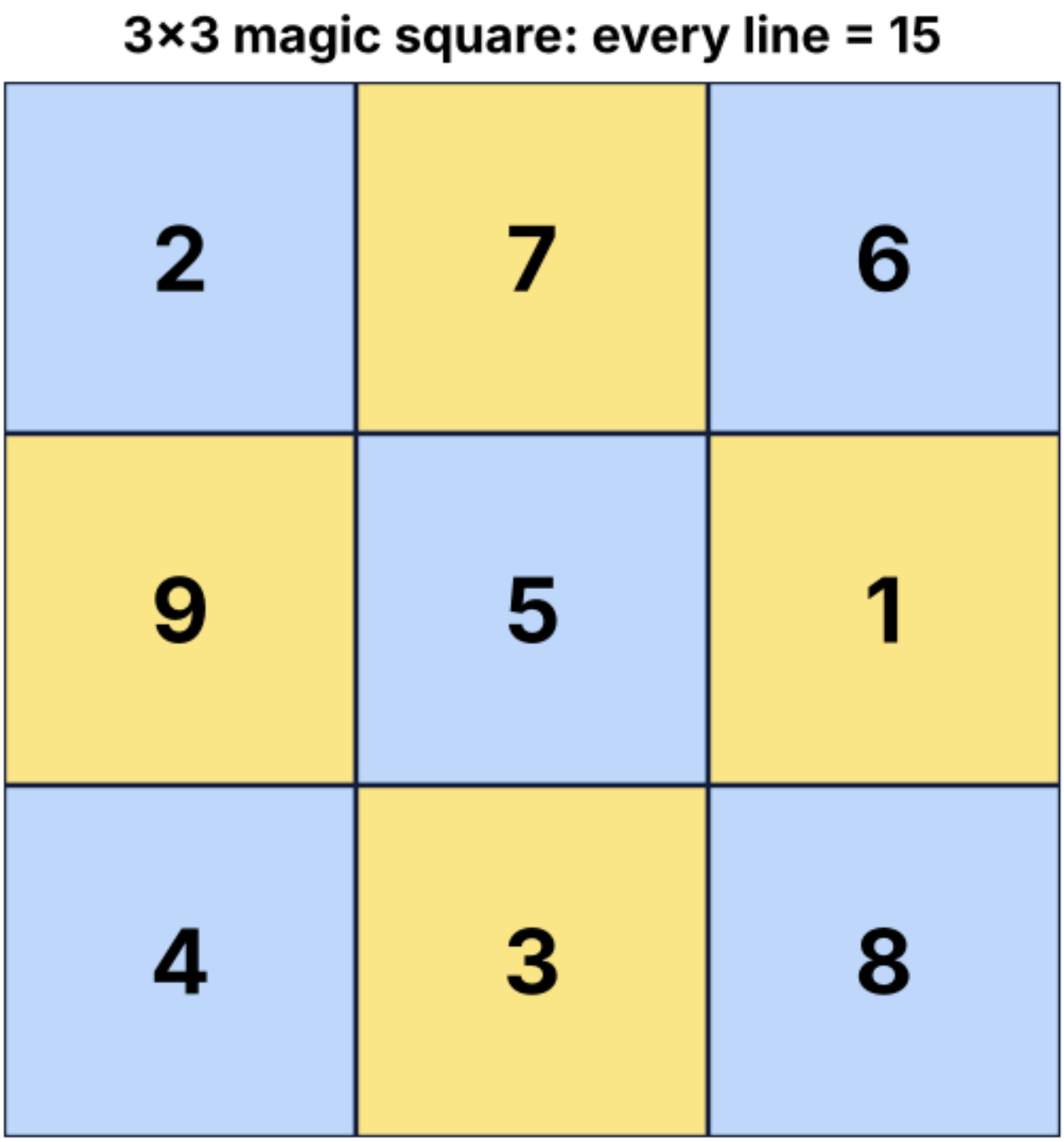
M = 1 because the sum of two 4-digit numbers is below 20,000. In the thousands column S + M (+ carry) must produce a 2-digit result, so S = 9 (or 8 with a carry) and O = 0. Following the columns gives the unique digits. A program tried all 1,814,400 digit assignments and found exactly one solution.

In TWO + TWO = FOUR (distinct digits, no leading zero), how many solutions exist, and what is the largest value of FOUR?

7 solutions; max FOUR = 1876

F is 1 (carry). T ≥ 5 so that 2T has a carry. A search over all digit assignments gave the count above.

Magic squares



A 3×3 magic square with magic sum 15.

How many different 3×3 magic squares can be made from the digits 1–9, and what is the centre cell?

8 squares (rotations/reflections of one); centre = 5

A program tested all 362,880 arrangements. Every one has 5 in the centre and the magic sum is 15 = 3 × 5.

Number-grid puzzles

Find the missing number: rows (3, 8, 24), (4, 5, 20), (6, 2, ?). Rule: the third is the product of the first two.

(3, 8, 24), (4, 5, 20), (6, 2, ?)

12

3 × 8 = 24, 4 × 5 = 20, so 6 × 2 = 12.

Find the missing number: 2, 6, 12, 20, 30, ?

42

Differences 4, 6, 8, 10, 12. Equivalently n(n+1): 1×2, 2×3, 3×4, 4×5, 5×6, 6×7 = 42.

Pattern	Looks like	Method
Cryptarithm	SEND + MORE = MONEY	Carry and column analysis
Magic square	All lines sum to S	Centre = S/3
Grid with product	(3, 8, 24)	Row relation
Digit conditions	"tens is double hundreds"	Equations then brute force
Missing figure	Triangle: 3, 5, 8	Sum, product or square
Number puzzle	Sum of digits and reverse	Algebra on digits

Shortcuts and tricks

- Use mod 9 (digit sum) to test a multiplication or addition quickly.
- In a cryptarithm, start with the letter that can only be 1 or 0.
- Magic square: centre = sum/3 and corners sum with the centre.

Common mistakes

- Letting two letters share a digit.
- Ignoring the carry.
- Starting at the units column when the leading column is more constrained.

Practice set: Number-puzzle practice

1. Missing: 3, 5, 9, 17, ? — Basic
2. In a magic square with sum 15, what is the centre? — Basic
3. Missing: 1, 4, 9, 16, 25, ? — Basic
4. How many 3-digit palindromes have digit sum 12? — Medium
5. How many two-digit numbers equal 4 times the sum of their digits? — Medium
6. Find the largest 3-digit number divisible by 9 whose digits are all different. — Medium

7.	In TWO + TWO = FOUR, what is the smallest value of FOUR?	Advanced
8.	How many ways can 6 be written as an ordered sum of three positive integers?	Advanced
9.	Sum of digits of all numbers from 1 to 99?	Advanced
10.	How many 4-digit numbers have all digits different?	Advanced

ADVANCED REASONING · SPATIAL AND TIME · IMPORTANT

11 Calendar, Clock, Cubes, Dice and Counting Figures

These questions look different but each uses one memorised result. Calendar: **odd days**. Clock: the **5.5 degrees per minute** rule. Cubes: counting corners, edges, faces. Dice: opposite faces. Figures: counting formulas for triangles, squares, rectangles.

Calendar

Odd-day method

- An ordinary year has 365 = 52 weeks + 1 odd day; a leap year has 2 odd days.
- **100 years = 5 odd days; 200 years = 3; 300 years = 1; 400 years = 0.**
- Days of a month in odd days: 31 → 3, 30 → 2, Feb → 0 (or 1 if leap).
- **Leap year:** divisible by 4; century years must be divisible by 400.
- Odd days → day: 0 Sunday, 1 Monday ... 6 Saturday (if counting from Sunday).
- Same calendar repeats after 28 years (inside a century) or when the first day and leap status match.

What day of the week is 15 August 2027?

Sunday

Count days from a known date: 1 January 2027 is a Friday. From 1 Jan to 15 Aug is 226 days = 32 weeks + 2 odd days, so the day is Sunday.

How many odd days are there in 400 years, and is 2100 a leap year?

0; not a leap year

400 years contain exactly 97 leap years: 400 × 365 + 97 = 146,097 days = 20,871 weeks. 2100 is a century not divisible by 400, so it is not leap.

If 1 January 2027 is a Friday, what day will 1 January 2028 be?

Saturday

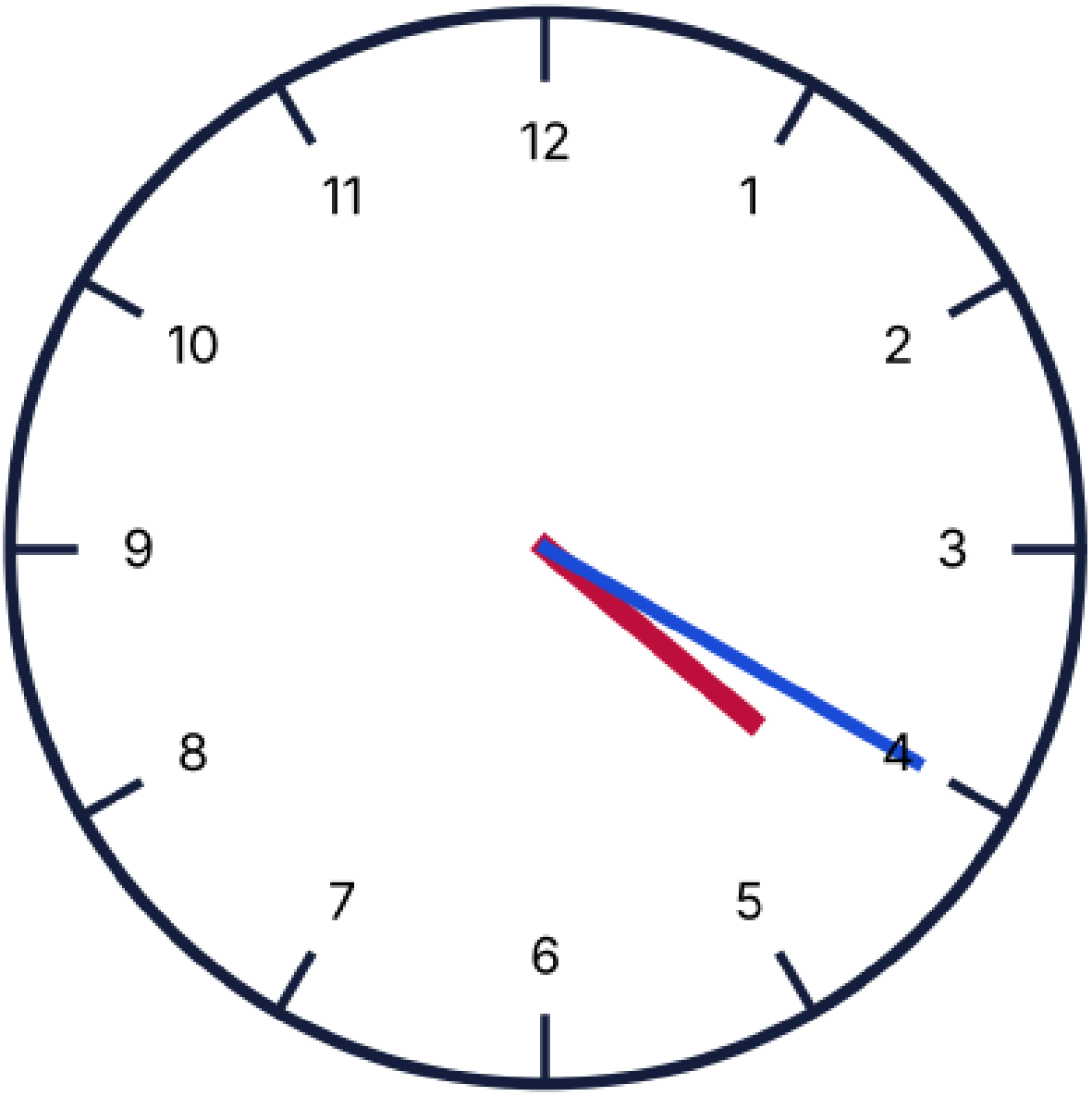
2027 has 365 days = 1 odd day, so Friday moves to Saturday.

How many Sundays are there in a leap year that begins on a Saturday?

53

2028 begins on Saturday. 366 = 52 weeks + 2 extra days (Saturday, Sunday), so 53 Sundays.

Clock



4:20 → angle = |30·4 − 5.5·20| = 10°

Minute hand moves 6° per minute, hour hand 0.5° per minute.

Clock formulas

- **Angle at h:m = |30h − 5.5m|** (take 360 − result if more than 180).
- Hands coincide every 65 5/11 minutes; in 12 hours 11 times; in 24 hours 22 times.
- Hands are opposite 11 times in 12 hours.
- Hands are at right angles 22 times in 12 hours (44 in 24).
- A clock that gains/loses x minutes per hour: compute the total shift over the elapsed time.

Find the angle between the hands at 4:20.

10

$|30\times4 - 5.5\times20| = |120 - 110| = 10°$.

At what time between 4 and 5 will the hands be together?

4:21 9/11 (that is 21.818 min past 4)

Hands coincide when 30h = 5.5m, so m = 120/5.5 = 21.818 minutes past 4.

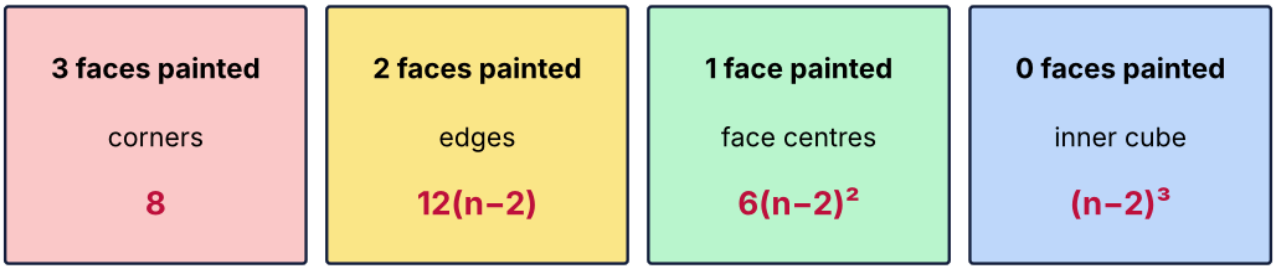
Angle between the hands at 9:15 and at 3:40?

172.5°; 130°

9:15: $|270 - 82.5| = 187.5 \rightarrow 360 - 187.5 = 172.5°$. 3:40: $|90 - 220| = 130°$.

Cubes

A painted cube of side n cut into n^3 unit cubes



Painted-cube counts for a cube of side n.

A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. How many cubes have (a) 3 faces, (b) 2 faces, (c) 1 face, (d) no face painted?

8; 24; 24; 8

Corners 8; edges $12(4-2) = 24$; face centres $6\times2^2 = 24$; inner $(4-2)^3 = 8$. Total $64 = 64$ ✓.

Dice

A standard die has opposite faces summing to 7 (1–6, 2–5, 3–4). Two views of the same die show the touching faces; a face not shown in either view is opposite to a shown one.

A die shows 1 on top, 2 on the front and 3 on the right. What is on the bottom, the back and the left?

6; 5; 4

Opposite faces sum to 7: bottom 6, back 5, left 4.

Counting figures

Figure	Count	Formula

Squares in n × n grid	30 (n=4)	Σk^2
Rectangles in 4 × 3 grid	60	C(5,2)·C(4,2)
Triangles, 4-line division	27	n(n+2)(2n+1)/8 (floor)
Diagonals of n-gon	n(n−3)/2	—
Handshakes among n	n(n−1)/2	—

Shortcuts and tricks

- For calendars, count weeks and remainder rather than months.
- For clock angles, draw the hands.
- Cube questions: use the corner/edge/face breakdown, then verify the total n^3.
- For overlapping rectangles use the formula C(a+1,2)·C(b+1,2).

Common mistakes

- Using 365 days for a leap year.
- Forgetting to take the smaller angle at the clock.
- Assuming the “opposite faces” sum to 7 on a non-standard die.

Practice set: Practice

1. Day of week of 26 January 2027? **Basic**
2. Angle at 6:00? **Basic**
3. How many unit cubes in a cube of side 5? **Basic**
4. Angle between the hands at 3:30? **Medium**
5. How many leap years between 2000 and 2100 inclusive? **Medium**
6. A cube of side 6 is painted and cut into unit cubes. How many have exactly one face painted? **Medium**
7. How many squares are there in a 5 × 5 grid? **Advanced**
8. How many rectangles in a 5 × 5 grid? **Advanced**
9. How many times do the hands of a clock overlap in 24 hours? **Advanced**
10. 2 September 2027: what day? **Advanced**

ADVANCED REASONING · PATTERNS · MUST-DO

12 Input–Output, Coding–Decoding and Series

Pattern questions reward a short checklist: for numbers check **difference, ratio, squares, cubes, primes**; for letters check **shift, reverse alphabet position, sum of values**; for input–output machines find the **rule of one step**, then apply it again.

Series types to recognise

Type	Example	Next
Arithmetic	4, 7, 10, 13	16
Geometric	3, 6, 12, 24	48
Squares / cubes	1, 8, 27, 64	125
Difference of differences	2, 3, 6, 11, 18	27
Alternating	2, 5, 4, 7, 6	9
Fibonacci type	1, 1, 2, 3, 5	8
$n^2 \pm n$	2, 6, 12, 20	30
Primes	2, 3, 5, 7	11

Recognise these eight patterns and most series are solved in seconds.

Coding rules

- **Letter value:** A=1 … Z=26. Opposite letter: A↔Z, B↔Y (sum 27).
- **Shift coding:** every letter moves +k. CAT → +1 → DBU.
- **Reverse coding:** write the word backward.
- **Number coding:** letters replaced by position values; compare words to find the value of each letter.
- **Condition coding:** a rule depends on whether a letter is a vowel / consonant.

If TCS is coded as UDT (+1 shift), how is NQT coded?

ORU

Every letter moves forward by 1: N→O, Q→R, T→U.

If the opposite-letter code is used (A↔Z, B↔Y …), how is CODE written?

XLWV

C↔X, O↔L, D↔W, E↔V.

If A = 1 … Z = 26, what is the value of the word TCS?

42

T = 20, C = 3, S = 19. Sum = 42.

In a code, GOD is 26 and DOG is also 26. What is the code for BAD if the code is the sum of letter positions?

7

B = 2, A = 1, D = 4 → 7.

Series

Next term of 4, 7, 10, 13, ?

16

Pattern: arithmetic.

Next term of 2, 3, 6, 11, 18, ?

27

Pattern: second difference constant.

Next term of 3, 6, 12, 24, ?

48

Pattern: geometric.

Next term: 1, 4, 9, 16, 25, ?

36

Squares of 1, 2, 3, 4, 5, 6.

Next term: 2, 6, 12, 20, 30, ?

42

n(n+1): the 6th term is 6 × 7.

Next letters: A, C, F, J, ?

O

Gaps increase: +2, +3, +4, +5. J + 5 = O.

Input–output machines

A machine rearranges a list step by step. Rule: in step 1 the **smallest number** is moved to the front; in step 2 the **alphabetically first word** is moved to the front; step 3 moves the next smallest number, and so on.

Step	Arrangement
Input	58 · pen · 12 · zoo · bat · 37
Step 1	12 · 58 · pen · zoo · bat · 37
Step 2	bat · 12 · 58 · pen · zoo · 37
Step 3	37 · bat · 12 · 58 · pen · zoo

What is the arrangement after step 3 of the machine above?

37 · bat · 12 · 58 · pen · zoo

Step 1 moves 12 to the front, step 2 moves “bat”, step 3 moves 37. Each step is applied to the previous result.

Type	Rule	First check
Number series	Difference	d_1, d_2, ratio
Letter series	Position gaps	A=1 ... Z=26
Coding	Shift / reverse / value	Compare two words
I/O machine	Simple sort rule	Do steps 1 and 2 to see the pattern
Odd one out	Common property	Parity, prime, square

Shortcuts and tricks

- For a long series, split the odd and even positions into two sequences.
- For letter series, convert to numbers.
- In input–output, never guess the rule from one step: use two.

Common mistakes

- Skipping the letter value check (J + 5 = O, not P).
- Using a pattern that fits only the first three terms.
- Applying the rule from the original input instead of the last step.

Practice set: Pattern practice

1. Next: 5, 10, 15, 20, ? — Basic

2. TCS shifted by +2: ? — Basic

3. Value of NQT (A=1) : — Basic

4. Next: 3, 9, 27, 81, ? — Medium

5. Next: 1, 2, 4, 7, 11, ? — Medium

6. Opposite-letter code of ATC? — Medium

7. Next: 2, 3, 5, 7, 11, 13, ? — Advanced

8. Next: 1, 1, 2, 3, 5, 8, 13, ? — Advanced

9. Next: 2, 5, 4, 7, 6, ? — Advanced

10. Next letter: B, D, G, K, ? — Advanced

13 ADVANCED REASONING · REVISION
Caselets, Mini Test and Final Sheet

Caselets mix several types: a short story with a few questions. The skill is to **decode the story once into a table** and answer every question from it. After the caselet comes a timed mini test and a one-page revision sheet.

Advanced Reasoning one-page sheet

Topic	One-line method
Seating	Fix one person; make blocks; verify.
Puzzle grid	Anchor first, negatives last.
Ranking	L + R − 1 = total.
Syllogism	Venn: accept only what holds in every diagram.
Inequality	Same direction links only.
Sufficiency	Test I, test II, then both.
Relations	Draw the tree; mark gender.
Directions	x, y coordinates; Pythagoras.
Liar puzzles	Try each person as the truth-teller.
Cryptarithm	Carry and leading digit first.
Calendar	Weeks + odd days.
Clock	$\lvert 30h - 5.5m \rvert$.
Cubes	$8 / 12(n-2) / 6(n-2)^2 / (n-2)^3$.
Series	d_1, d_2, ratio, squares.

Keep this table for the last day.

Caselet: six employees, three departments

Six employees A, B, C, D, E, F work in three departments: HR, IT and Sales. Exactly two employees work in each department. A and B are in different departments; C and D are in the same department; E and F are in different departments; A is in HR; C is not in HR; B is in IT; E is not in IT.

Who works in Sales, and where does F work?

C, D; F in IT

Tabulate: A = HR, B = IT. C and D are together and C is not in HR, so they are in IT or Sales. IT already has B, so C and D cannot both fit in IT; they form the Sales pair. HR then needs one more and IT one more among E, F. E is not in IT, so E = HR and F = IT. A program checked all 90 ways to place two employees in each department; exactly one satisfies every clue.

Who works with A?

E

A is in HR; the other HR employee is E.

Timed mini test: 12 questions, 25 minutes

Practice set: Mini test (set a 25-minute timer)

1.	Five people sit in a row. How many ways if A and B must sit together?	Basic
2.	X is 14th from the left and 22nd from the right. Total?	Basic
3.	P > Q ≥ R = S. Is P > S true / false / uncertain?	Basic
4.	All X are Y. All Y are Z. Does "All Z are X" follow?	Medium
5.	How many matches in a 33-team knockout?	Medium
6.	A man walks 9 m east and 12 m north. Distance from start?	Medium
7.	Angle between clock hands at 7:20?	Medium
8.	Next: 2, 6, 12, 20, 30, ?	Medium
9.	Which day is 1 January 2028 if 1 January 2027 is Friday?	Medium
10.	A cube of side 5 is painted and cut into 1 cm cubes. How many have no painted face?	Advanced
11.	How many different 3-digit numbers can be made from 1–5 without repetition?	Advanced
12.	Trailing logic: "If P then Q". Not Q. What follows?	Advanced

Shortcuts and tricks

- Score 10+ : exam-ready; 7–9: revise weak chapters; below 7: redo chapters 2–6 first.
- Review every wrong answer: mark whether the mistake was reading, method or arithmetic.

Answer key

Answers to every practice set. B = Basic, M = Medium, A = Advanced.

1. Strategy

1. C (B)
2. I (B)
3. 120 (B)

2. Seating

1. 120 (B)
2. 120 (B)
3. seat 7 (B)
4. 36 (M)
5. 144 (M)
6. 48 (M)
7. 240 (M)
8. 720 (A)
9. F (A)
10. 72 (A)

3. Multi-attribute puzzles

1. 15 (B)
2. 4 (B)
3. S (B)
4. A (M)
5. B (M)
6. 2 (M)
7. 3 (M)
8. S (A)
9. 120 (A)
10. 1 (A)

4. Ranking and tournaments

1. 13 (B)
2. 63 (B)
3. 45 (B)
4. 27 (M)
5. 10 (M)
6. 99 (M)
7. 40 (M)
8. 17 (A)
9. 16 (A)
10. 10 (A)

5. Syllogism

1. does not follow (B)
2. follows (B)
3. follows (M)
4. does not follow (M)
5. follows (M)
6. does not follow (M)
7. follows (A)
8. follows (A)
9. does not follow (A)
10. does not follow (A)

6. Inequalities

1. true (B)
2. uncertain (B)
3. uncertain (B)
4. true (M)
5. true (M)
6. true (M)
7. uncertain (A)
8. uncertain (A)
9. uncertain (A)
10. true (A)

7. Data sufficiency

1. Statement I alone is sufficient (B)
2. Both together are needed (B)
3. Both together are needed (M)
4. Even both together are not sufficient (M)
5. Both together are needed (M)
6. Both together are needed (M)
7. Statement II alone is sufficient (A)
8. Both together are needed (A)
9. Both together are needed (A)
10. Both together are needed (A)

8. Relations and Directions

1. daughter (B)
2. aunt (B)
3. South (B)
4. cousin (M)
5. uncle (M)
6. aunt (M)
7. 10.0 (M)
8. 4 m west (A)
9. uncle (A)
10. grandmother (A)

9. Truth-tellers and critical reasoning

1. It did not rain (B)
2. No (B)
3. Yes (B)
4. 1 (exactly one) (M)
5. A, B (M)
6. Yes (M)
7. 32 (A)
8. 3 (A)
9. If A then B (A)
10. If not B then A (A)

10. Cryptarithm and number puzzles

1. 33 (B)
2. 5 (B)
3. 36 (B)
4. 5 (M)
5. 4 (M)
6. 981 (M)
7. 1468 (A)
8. 10 (A)
9. 900 (A)
10. 4536 (A)

11. Calendar, clock, cubes

1. Tuesday (B)
2. 180 (B)
3. 125 (B)
4. 75 (M)
5. 25 (M)
6. 96 (M)
7. 55 (A)
8. 225 (A)
9. 22 (A)
10. Thursday (A)

12. Input–output, coding, series

1. 25 (B)
2. VEU (B)
3. 51 (B)
4. 243 (M)
5. 16 (M)
6. ZGX (M)
7. 17 (A)
8. 21 (A)
9. 9 (A)
10. P (A)

13. Caselets and mini test

1. 48 (B)
2. 35 (B)
3. true (B)
4. does not follow (M)
5. 32 (M)
6. 15 (M)
7. 100 (M)
8. 42 (M)
9. Saturday (M)
10. 27 (A)
11. 60 (A)
12. Not P (A)